Compassion International

Scaling Global Communication Infrastructure for 150M+ Messages

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At Compassion International, our global communication pipeline, tied to $1B+ in annual donor transactions, was bottlenecked at roughly 13 million messages a year, threatening our outreach schedules when paired with downstream XMPie engines.

As part of a targeted engineering task force, I re-architected the delivery pipeline to scale capacity past 150 million messages annually without dropping transactions. I co-designed a stateful C# Windows queuing service that buffered, batched, and precisely polled payloads to prevent overwhelming downstream systems. To handle the massive data prep, I authored complex MS SQL Server data pipelines utilizing dynamic T-SQL pivots, unpivots, and staging tables.

Impact: We successfully hit our 150M+ capacity target. Furthermore, the lightweight, REST-based error logging system I built for resilience across distributed components performed so reliably that enterprise leadership adopted it as the mandatory logging standard across all subsequent projects.